JOONWON CHOI

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Research Goal

Develop reliable autonomy for complex, interactive systems by modeling sophisticated dynamics and interactions; formally analyzing safety; and generating adaptive, resilient decisions

Aeras of Interest

►Safety (reachability) analysis ►Koopman operator ►Control theory ►Multi-Agent System (MAS) ►Human Autonomy Teaming **(HAT)** ►Human-Robot Interaction (HRI) ►Continual reinforcement learning ►Formal methods

Education

➢ **Purdue University** West Lafayette, IN

Ph.D., School of Aeronautics and Astronautics May 2026

Specialization: HAT, Human-in-the-loop, Reachability analysis, Koopman operator, Formal method, Continual reinforcement learning, Simulation (Airsim, Isaac Sim, Isaac Lab)
Advisor: Ph.D. Inseok Hwang

➢ **Ulsan National Institute of Science and Technology** Ulsan, South Korea

M.S., Department of Mechanical Engineering Aug 2021

Specialization: Multi-agent system (Formation control, flocking, network connectivity preservation, aerial defense, interception), Autonomous system (ROS, flight experiment)
Advisor: Ph.D. Hyondong Oh

B.S., Department of Mechanical & Aerospace Engineering Mar 2019

Employment

➢ **Apollo 11 Postdoctoral Fellow,** Purdue University July 2026 - Present

My research interest is in *Safe and Resilient Teaming with Human-Aware Autonomy,* which aims for an autonomous agent that can truly act as a teammate, not merely a tool for automation. I mainly focus on the following three thrusts [Link]:

- *Human-aware autonomy teammate*: to develop autonomy that can accurately account for human decision-making processes and control policies affected by human cognitive states
- *LLM-based transparent interaction*: to establish an architecture that leverages LLMs for transparent and formally verifiable interaction for HAT
- *Safe and resilient decision-making of AI*: to design a novel architecture for human-aware autonomy that transparently communicates with humans via LLMs and understands humans through data-driven models, thereby truly acting as a teammate of humans with a safe and resilient decision-making process.

➢ **Bilsland Dissertation Fellow,** Purdue University May 2025 - May 2026

Secured last year fellowship for Ph.D. degree to develop (data-driven) safety analysis algorithm for human-aware systems using reachability analysis, formal method, and Koopman operator

–*Formal method-based safe HRI using LLM*: Designed a reliable HRI interface using LLMs, signal temporal logic (STL), and reachability analysis that automatically analyzes the feasibility & safety of human natural language commands and ensures safe operation
- *Continual Reinforcement Learning (CRL)*: Applied Koopman AutoEncoder (KAE) and Koopman decomposition techniques to facilitate CRL

➢ **Graduate Teaching Assistant,** Purdue University Aug 2024 – Dec 2024

Served as a TA for *AAE 668 & ECE 695 Hybrid Systems: Theory and Applications*, a graduate-level course focused on theoretical analysis & application of (linear) switching system, reachability analysis, stability analysis, and hybrid state estimation techniques

➢ **Graduate Research Assistant,** Purdue University Aug 2021 - May 2025

Participated in NSF funded project *CPS Frontier: Cognitive Autonomy for Human CPS: Turning Novices into Experts* to develop cognitive-aware cyber-physical-human-systems (CPHS) [Link]

- *Data-driven reachability analysis*: Designed data-driven (closed-loop) reachability analysis algorithms that can explicitly consider humans' control decisions to analyze the safety of a CPHS
- *Koopman Operator*: Analyzed control theoretical characteristics (controllability) of Koopman operator to verify the feasibility of the Koopman operator as a data-driven model for CPHS
– *Cognitive state estimator for human-aware HAT*: Designed a human workload estimator using various physiological & behavioral sensors (fNIRS, heart rate, camera) for adaptive task allocation of human-autonomy team. I also constructed a combined indoor platform; and 3D simulation environment using Python and Airsim; for human subject experiment

➢ **Researcher,** UNIST Mar 2019 – Mar 2020

Developed MPC-based multi-to-multi interception (defense) algorithm; and sensor fusion algorithm for vision-based autonomous landing quadrotor in maritime environment

- *Multi-Agent System*: Developed and proved the stability of algorithms for flocking, formation control, and network connectivity preservation
- *Hardware Experiment*: Participated in developing a sensor-fusion algorithm and ROS implementation for a vision-based autonomous ship landing system

Awards & Honors

1. **Apollo 11 Postdoctoral Fellowships** (July 2026 – May 2028)
2. Research Pitch Award, *Institute for Control, Optimization and Networks (ICON), Purdue University* (Jan 2026)
3. **Bilsland Dissertation Fellowship** (May 2025 – May 2026)
4. Excellence Award (Co-author), *Korea Aerospace Industries* (Oct 2021)
5. 2nd Prize in Team Challenge, 2nd Autonomous Drone Technology Challenge, *Agency for Defense Development* (Oct 2021)

6. **Korean Government Scholarship Program for Study Overseas** (Aug 2021)
7. Excellent Presentation Award, *Korean Society for Aeronautical & Space Sciences* (Jul 2021)
8. Excellent Presentation Award, *Korean Society for Aeronautical & Space Sciences* (Nov 2019)
9. Undergraduate Thesis Award, *Korea Robotics Society* (Jan 2019)

Skills

Programming: Python, Pytorch, and MATLAB; experience with C++

Packages & Experiment: experience with IsaacSim/Lab, ROS, Gazebo, Pixhawk/PX4, Motion capture system, Raspberry Pi, TX2

Research Experience

➢ **HUMAN AUTONOMY TEAMING (HAT) & DATA-DRIVEN APPLICATION**

- Participated in **NSF-funded project** (*CPS Frontier: Cognitive Autonomy for Human CPS: Turning Novices into Experts*) to develop cognitive-aware human-in-the-loop systems
- Collaborated to create a reliable human-robot interaction interface using Large Language Models (**LLMs**), **signal temporal logic (STL),** and **reachability analysis** that automatically analyzes the feasibility & safety of human natural language command and ensures safe operation
- Built 3D simulation environment using **Python** and **Airsim** for faster pilot training in urban UAM flight scenarios by providing adaptive flight assistance based on pilot skill level
- Utilized transformer to estimate human workload using various physiological & behavioral sensors (fNIRS, heart rate, camera) for adaptive task allocation of human-autonomy team
- Designed **data-driven reachability analysis** algorithm that can explicitly consider human's control decision to analyze the safety of a system driven by a human
- Constructed the distraction-aware safety verification framework that can capture human distraction using a fNIRS sensor to analyze the safety of a system driven by a human
- Used Gaussian Mixture Model (**GMM**) to generate data-driven human behavior model for predicting future human control decision based on imitation learning framework

➢ **MULTI-AGENT SYSTEM**

- Leveraged **MPC**-based impact time control guidance and genetic algorithm to realize efficient **multi-to-multi aerial defense (interception)** algorithm against adversarial swarm
- Designed a **formation control** algorithm for decentralized multi-agent systems that guarantees collision avoidance and network connection (Laplacian connectivity) preservation
- Joined in developing parameter optimization technique motivated by an ant colony in nature to obtain efficient emergent behavior of a **flock (Cucker-Smale model)**
- Developing a multi-agent deceptive planning algorithm that leverages unresolved measurements of radar to deceive the destination and cardinality of attacker drones

➢ **KOOPMAN OPERATOR & CONTINUAL REINFORCEMENT LEARNING**

- Analyzed control theoretical characteristics (controllability) of Koopman operator to verify the feasibility of Koopman operator as a data-driven model for CPHS
- Devising Koopman AutoEncoder (**KAE**) that leverages **reachability analysis** and **conformal prediction** to learn robust Koopman operator to enhance generalizability in unseen situations
- Constructed IsaacLab-based RL simulation platform to train KAE for several systems; including quadruped robot and crazyflie
- Applied KAE and Koopman decomposition techniques to improve **continual reinforcement learning (CRL)** by leveraging **policy reuse**
- Designing reversible LLM functionality extraction algorithm using KAE that can separate a desired skill of LLM and easily recover the original network

HARDWARE EXPERIMENT

- Constructed combined indoor **human subject experiment platform** with a multi-rotor vehicle, motion capture system, fNIRS, heart rate sensor, and vision sensor
- Testified proposed LLM-based reliable HAT framework via **indoor flight experiment** using crazyflie to solve signal temporal logic (**STL**) constrained MILP
- Implemented feedforward Image-Based Visual Servoing (**IBVS**) and **Kalman filter** for vision-based landing of a multi-rotor vehicle on a moving ship in severe maritime conditions
- Participated in several field landing experiments using customized vehicle equipped **Pixhawk, camera, and Raspberry PI/TX2/Xavier**
- Developed a multi-rotor operated by YOLO with pruning, visual SLAM using camera sensor, and Path planning with collision avoidance (RRT, DWA) for an autonomous drone competition
- Familiarized in validation through simulation and experiment with **ROS1, Gazebo, PX4**

Activities

MENTORING & TEACHING

- Collaborated with graduate student mentors and faculty to facilitate the Summer Intensive Research Internship (SIRI) program over multiple years
- Served as a **teaching assistant** (*AAE 668 & ECE 695 Hybrid Systems: Theory and Applications*)
- Supervised and mentored multiple master's and undergraduate research interns
- Took specialized class for teaching assistantship *ENGL 62000: Classroom Communication In ESL*
- Participated in *Future Faculty Boot Camp (FFBC)* which takes place at Purdue University
- Participating in Purdue Institute for Control, Optimization and Networks (**ICON**) Control Reading Group since Summer 2024

SERIVCES

- Journal Papers Reviewer
  - IEEE Transactions on Aerospace and Electronic Systems (TAES)
- Conference Papers Reviewer

- 2024 American Control Conference (ACC)
- 2024 Conference on Decision and Control (CDC)
- 5th IFAC Workshop on Cyber-Physical and Human Systems (CPHS 2024)
- 2025 Conference on Decision and Control (CDC)
- 23rd International Federation of Automatic Control World Congress (IFAC)
- 2026 Conference on Decision and Control (CDC)

Publications

A total of 8 Journal and 8 Conference papers have been ***Published***; 1 Journal paper is ***Under Review;*** 3 Journal papers are ***In Preparation***

Journals

1. J. Choi, J. Hunter, N. Jain, I. Hwang, and T. Reid. "Distraction-aware Data-driven Hybrid Reachability Analysis of Cyber-Physical-Human Systems Using Functional Near-infrared Spectroscopy." *Journal of Aerospace Information Systems* (**Under Review**)
2. J. Choi, S. Byeon, and I. Hwang. "Data-Driven Forward Stochastic Reachability Analysis for Nonlinear Human-in-the-Loop Systems." *IEEE Transactions on Control Systems Technology* (2024)
3. J. Choi, H. Park, and I. Hwang. "Bootstrapped Gaussian Mixture Model-Based Data-Driven Forward Stochastic Reachability Analysis." *IEEE Control Systems Letters with ACC option* (2023)
4. J. Choi, M. Seo, H. Shin, and H. Oh. "Adversarial Swarm Defence Using Multiple Fixed-Wing Unmanned Aerial Vehicles." *IEEE Transactions on Aerospace and Electronic Systems* 58.6 (2022): 5204-5219.
5. J. Choi, Y. Song, S. Lim, and H. Oh. "Decentralized multi-subgroup formation control with connectivity preservation and collision avoidance." *IEEE Access* 8 (2020): 71525-71534.
6. S. Byeon, J. Choi, Y. Zhang, and I. Hwang, "Stochastic-Skill-Level-Based Shared Control for Human Training in Urban Air Mobility Scenario," *ACM Transaction on Human-Robot Interaction*, (2024)
7. S. Byeon, J. Choi, and I. Hwang, "A Computational Framework for Optimal Adaptive Function Allocation in a Human-Autonomy Teaming Scenario." *IEEE Open Journal of Control Systems* (2023).
8. G. Cho, J. Choi, G. Bae, and H. Oh. "Autonomous ship deck landing of a quadrotor UAV using feed-forward image-based visual servoing." *Aerospace Science and Technology* 130 (2022): 107869.
9. Y. Song, M. Gu, J. Choi, H. Oh, S. Lim, H. Shin, A. Tsourdos. "Using lazy agents to improve the flocking efficiency of multiple UAVs." *Journal of Intelligent & Robotic Systems* 103 (2021): 1-17.

Conferences

1. J. Choi, J. Hunter, I. Hwang, and T. Reid. " Distraction-aware Data-driven Hybrid Reachability Analysis of Cyber-Physical-Human Systems Using Functional Near-infrared Spectroscopy." *2025 American Institute of Aeronautics and Astronautics (AIAA) SciTech*.
2. J. Choi, M. Cho, H. Park, V. Vishnu, and I. Hwang, "On the Controllability Preservation of Koopman Bilinear Surrogate Model," *63rd IEEE Conference on Decision and Control (CDC 2024)*.
3. J. Choi, S. Byeon, and I. Hwang. "Data-Driven Forward Stochastic Reachability Analysis for Human-In-The-Loop Systems." *62nd IEEE Conference on Decision and Control (CDC 2023)*, IEEE, 2023.

4. J. Choi, S. Byeon, and I. Hwang. "State Prediction of Human-in-the-Loop Multi-rotor System with Stochastic Human Behavior Model." *IFAC-PapersOnLine* 55.41 (2022): 119-124.
5. S. Byeon, M. Yuh, J. Choi, N. Jain, and I. Hwang, "Function Allocation Assessment in Human-Autonomy Teaming Using Workload Metrics from Heart Rate Sensor." *2025 American Institute of Aeronautics and Astronautics (AIAA) SciTech*.
6. C. Yang, J. Choi, S. Anandavel, and I. Hwang. "Reachability Analysis of Human-in-the-Loop Systems Using Gaussian Mixture Model with Side Information." *2024 IEEE International Conference on Advanced Intelligent Mechatronics (AIM)*. IEEE, 2024.
7. Y. Song, J. Choi, H. Oh, M. Lee, S. Lim, and J. Lee. "Improvement of decentralized flocking flight efficiency of fixed-wing UAVs using inactive agents." *AIAA SciTech 2019 Forum*. 2019.
8. S. Lim, Y. Song, J. Choi, H. Myung, H. Lim, and H. Oh. "Decentralized hybrid flocking guidance for a swarm of small UAVs." *2019 Workshop on Research, Education and Development of Unmanned Aerial Systems (RED UAS)*. IEEE, 2019.

Preprints

1. J.Choi, K. Pant, K. Nune, and I.Hwang, "Reachability-based Temporal Logic Verification for Reliable LLM-guided Human-Autonomy Teaming." Preprint: https://arxiv.org/abs/2603.08633
2. J.Choi, K.Pant, Y. Nam, H. Hellmann, K. Nune, and I.Hwang, "Signal Temporal Logic Verification and Synthesis Using Deep Reachability Analysis and Layered Control Architecture," Preprint: https://arxiv.org/abs/2602.23313

In Preparation

1. J.Choi, Y.Guo, and I.Hwang, "Koopman Modularization: From Decomposition to Reconstruction." targeting for *IEEE Transactions on Robotics (T-RO)*.
2. J.Choi and I.Hwang, "Conformal Koopman Autoencoder for Learning Reachability-Informed Koopman Operator." targeting for *IEEE Transactions on Neural Networks and Learning Systems (TNNLS)*.
3. J.Choi, S.Byeon, Y.Nam, I.Hwang, and C.Kwon "Multi-agent Intent-deceptive Planning Using Unresolved Measurement," targeting for *IEEE Control Systems Letters (L-CSS)*.

Posters & Talks

1. J. Choi, Y. Guo, and I. Hwang, "Koopman Operator-based Network Modularization for Policy Reuse" *Institute for Control, Optimization and Networks (ICON) Student Research*, 2026. ***Presenter:*** *J. Choi*
2. J. Choi, K. Pant, K. Nune, and I. Hwang, "Verification of temporal logic specification using reachability for reliable LLM-guided human-robot interaction" *Purdue Applied AI Research Center (AARC) Student Poster Session at Applied AI Day*, 2025. ***Presenter:*** *J. Choi*
3. S. Byeon, J. Choi, and I. Hwang, "Cognitive-Aware Multi-Modal Human-Swarm Interface for Optimal Collaboration." *Revolutionizing Autonomous Systems with Generative AI, and Large Language Models for Zero Trust Architecture: Enhancing Security, Resilience, and Safety of Systems Managing Edge Autonomous Devices (GENZERO24),* 2024. ***Presenter:*** *J. Choi*